TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	82.9 ± 1.76
Correlation Stability	89.7 ± 2.19
Distribution Consistency	89.6 ± 1.69
Entropy Preservation	89.9 ± 1.37
Mean Z Score Drift	0.1 ± 0.03
Categorical Fidelity	84.5 ± 1.02
Privacy Risk	48.2 ± 0.75
Data Density	99.8 ± 0.28
Coverage Score	32.1 ± 4.48
Feature Accuracy	85.0 ± 4.73
Novelty Score	0.5 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	43.547	-1.199	2.7%	OK
TB	3.299	2.774	-0.524	15.9%	Warn
DB	1.486	1.170	-0.317	21.3%	Warn
Alkphos	290.576	276.494	-14.083	4.8%	OK
Sgpt	80.714	52.322	-28.392	35.2%	Drift
Sgot	109.911	88.337	-21.574	19.6%	Warn
TP	6.483	6.668	+0.184	2.8%	OK
ALB	3.142	3.342	+0.200	6.4%	OK
A/G Ratio	0.947	0.941	-0.006	0.6%	OK
Selector	1.286	1.311	+0.025	1.9%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	82.9 ± 1.76
Correlation Stability (*2)	89.7 ± 2.19
Distribution Consistency (*3)	89.6 ± 1.69
Entropy Preservation (*4)	89.9 ± 1.37
Mean Z Score Drift (*5)	0.1 ± 0.03
Categorical Fidelity (*6)	84.5 ± 1.02
Privacy Risk (*7)	48.2 ± 0.75
Data Density (*8)	99.8 ± 0.28
Coverage Score (*9)	32.1 ± 4.48

Feature Accuracy (*10)	85.0 ± 4.73
Novelty Score (*11)	0.5 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6837606837606838	0.6	0.6982905982905983	0.014529914529914511	
F1	0.7893381394683538	0.7067700517756083	0.7900970606233839	0.00075892115503007 89	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6222222222222223	0.6803418803418804	-0.005982905982905973	
F1	0.7939840886612185	0.733325234242667	0.7780961884002601	-0.015879002609584 32	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6991452991452991	0.7068376068376068	0	
F1	0.8279091624529537	0.7943419127449458	0.8043347269382195	-0.02357443551473402 7	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6581196581196582	0.5897435897435898	0.6760683760683761	0.01794871794871795	
F1	0.7628098669733415	0.6993750670764503	0.7700480193204683	0.00723815234712703 6	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.6572649572649573	0.7068376068376068	0.00085470085470086 27	
F1	0.8128002697439015	0.7620083012892702	0.7994341443136168	-0.01336612543028462 5	

Model: NB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.5854700854700854	0.5376068376068377	-0.02136752136752136	
F1	0.5587432119380917	0.6600296747707224	0.5301511125271533	-0.02859209941093831	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6145299145299147	0.6871794871794872	-0.01538461538461538	
F1	0.7997903571549948	0.723843403171767	0.780960831440232	-0.01882952571476270 7	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6393162393162393	0.6777777777777778	0.00341880341880341 8	
F1	0.7821343746683586	0.7509068853389722	0.7827528735221019	0.00061849885374326 56	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7111111111111111	0.6555555555555557	0.7119658119658119	0.00085470085470086 27	
F1	0.8140261582907407	0.7599282264928531	0.8048023023118086	-0.00922385597893260 5	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6743589743589744	0.6974358974358974	0.01111111111111117	
F1	0.7971299319768493	0.7699525520779424	0.7923497678573452	-0.00478016411950401 6	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	61	Male	0.8	0.2	163	18	19	6.3	2.8	0.8
2	54	Male	0.8	0.2	218	20	19	6.3	2.5	0.6
3	65	Male	1.1	0.3	258	48	40	7.0	3.9	1.2
4	60	Male	2.3	0.6	272	79	51	6.6	3.5	1.1
5	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
6	21	Male	0.8	0.2	183	33	57	6.8	3.5	1.0
7	29	Female	1.0	0.3	160	22	30	6.8	3.5	1.2
8	47	Male	2.1	0.5	210	35	62	7.0	3.6	0.85
9	34	Male	0.9	0.2	180	29	44	6.5	3.2	1.07
10	56	Female	1.5	0.7	240	40	50	6.7	2.8	0.75
11	43	Male	3.2	1.5	290	60	100	7.1	3.4	0.91
12	38	Female	0.8	0.1	150	12	22	5.9	3.0	1.26
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	32	Male	30.5	17.1	218	39	79	5.5	2.7	0.9
2	50	Female	1.7	0.6	430	28	32	6.8	3.5	1.0
3	22	Male	0.6	0.2	202	78	41	8.0	3.9	0.9
4	33	Male	0.8	0.2	198	26	23	8.0	4.0	1.0
5	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
6	48	Female	0.8	0.2	150	25	23	7.5	3.9	1.0
7	29	Female	0.6	0.2	155	24	30	6.4	3.4	1.0
8	67	Male	5.4	2.8	310	50	60	6.1	2.8	0.9
9	21	Male	1.1	0.4	140	27	35	5.8	2.7	1.1
10	37	Female	0.8	0.3	220	30	30	7.0	3.4	1.25
11	45	Male	1.2	0.5	185	42	51	6.5	3.1	0.95
12	52	Female	0.9	0.2	170	32	38	6.8	3.5	0.85
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Male	0.8	0.2	208	25	50	7.1	3.7	1.0
2	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8
3	18	Male	0.6	0.2	538	33	34	7.5	3.2	0.7
4	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
5	31	Male	1.3	0.5	184	29	32	6.8	3.4	1.0
6	61	Male	0.8	0.2	192	28	35	6.9	3.4	0.9
7	29	Female	1.1	0.4	210	30	35	7.5	4.0	1.2
8	37	Male	0.9	0.3	190	55	70	6.9	3.3	0.9
9	49	Female	0.8	0.1	150	28	30	6.0	3.5	1.1
10	53	Male	1.5	0.8	300	85	90	6.8	3.0	0.6
11	22	Male	0.6	0.2	170	20	25	7.0	4.2	1.2
12	62	Female	2.3	1.1	320	42	33	7.1	2.8	0.7
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
2	50	Male	0.9	0.3	901	23	17	6.2	3.5	1.2
3	42	Female	0.7	0.2	152	35	81	6.2	3.2	1.06
4	33	Male	1.8	0.8	196	25	22	8.0	4.0	1.0
5	60	Male	2.9	1.3	230	32	44	5.6	2.0	0.5
6	42	Male	8.9	4.5	272	31	61	5.8	2.0	0.5
7	34	Female	1.2	0.4	180	30	70	7.2	4.0	1.1
8	45	Male	0.8	0.2	210	55	65	6.9	3.5	0.88
9	22	Female	0.7	0.3	140	25	20	7.0	4.2	1.2
10	39	Male	3.5	1.2	300	45	100	6.5	2.8	0.84
11	50	Female	0.9	0.2	220	40	50	6.8	3.3	1.3
12	56	Male	1.3	0.4	160	32	75	7.1	3.0	0.95
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	57	Male	0.7	0.2	208	35	97	5.1	2.1	0.7
2	60	Male	2.0	1.1	664	52	104	6.0	2.1	0.53
3	45	Male	2.2	1.6	320	37	48	6.8	3.4	1.0
4	21	Male	3.9	1.8	150	36	27	6.8	3.9	1.34
5	58	Male	1.7	0.8	188	60	84	5.9	3.5	1.4
6	28	Male	0.6	0.2	159	15	16	7.0	3.5	1.0
7	27	Male	1.5	0.6	175	32	50	6.2	3.1	0.87
8	54	Female	0.8	0.2	210	22	30	7.0	3.5	1.0
9	37	Female	2.1	1.0	220	45	60	6.5	3.0	0.67
10	22	Male	0.9	0.3	130	24	34	6.8	2.9	1.0
11	48	Male	3.7	1.5	300	65	60	7.2	3.7	1.2
12	63	Female	1.2	0.5	148	30	45	7.5	3.2	0.75
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	36	Female	0.8	0.2	650	70	138	6.6	3.1	0.8
2	36	Female	1.2	0.4	358	160	90	8.3	4.4	1.1
3	46	Male	0.8	0.2	160	31	40	7.3	3.8	1.1
4	37	Male	0.8	0.2	147	27	46	5.0	2.5	1.0
5	6	Male	0.6	0.1	289	38	30	4.8	2.0	0.7
6	21	Male	1.0	0.3	142	27	21	6.4	3.5	1.2
7	28	Female	0.9	0.3	250	30	45	7.0	3.5	1.0
8	34	Male	1.5	0.7	300	36	90	6.8	2.9	2.0
9	45	Female	0.7	0.2	170	40	54	5.8	3.0	1.0
10	50	Male	1.1	0.3	220	28	50	6.7	3.7	2.0
11	60	Female	0.8	0.1	190	25	15	7.1	3.9	1.0
12	29	Male	2.3	1.0	310	42	60	5.6	2.6	1.0
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	5.2	2.4	168	126	202	6.8	2.9	0.7
2	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
3	74	Male	0.6	0.1	272	24	98	5.0	2.0	0.6
4	22	Male	0.8	0.2	198	20	26	6.8	3.9	1.3
5	52	Male	2.7	1.4	251	20	40	6.0	1.7	0.39
6	34	Male	4.1	2.0	289	875	731	5.0	2.7	1.1
7	25	Female	1.0	0.4	210	30	45	6.5	3.0	1.2
8	40	Male	2.5	1.0	300	50	70	7.0	3.5	1.3
9	55	Female	3.1	1.2	250	40	60	6.8	3.2	1.0
10	30	Male	1.2	0.5	180	35	58	7.5	3.8	1.4
11	65	Female	0.5	0.2	150	20	30	5.9	2.8	0.9
12	45	Male	4.0	2.0	290	90	110	8.1	3.4	0.6
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	4	Male	0.9	0.2	348	30	34	8.0	4.0	1.0
2	34	Male	4.1	2.0	289	875	731	5.0	2.7	1.1
3	40	Male	0.7	0.1	202	37	29	5.0	2.6	1.0
4	40	Male	0.6	0.1	98	35	31	6.0	3.2	1.1
5	65	Male	0.7	0.2	265	30	28	5.2	1.8	0.52
6	50	Female	1.7	0.6	430	28	32	6.8	3.5	1.0
7	25	Female	1.0	0.3	210	22	31	6.8	3.5	0.9
8	39	Male	0.8	0.2	190	30	25	7.5	4.0	1.1
9	54	Male	2.5	1.0	400	44	70	5.8	2.9	0.6
10	48	Female	1.1	0.4	250	29	39	6.1	3.2	1.0
11	31	Male	0.7	0.1	300	22	34	8.0	4.1	1.2
12	67	Female	3.1	1.5	480	35	45	6.7	3.3	1.3
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	23	Female	2.3	0.8	509	28	44	6.9	2.9	0.7
2	70	Male	1.4	0.6	146	12	24	6.2	3.8	1.58
3	53	Female	0.9	0.2	210	35	32	8.0	3.9	0.9
4	40	Male	0.6	0.1	98	35	31	6.0	3.2	1.1
5	38	Male	0.7	0.2	216	349	105	7.0	3.5	1.0
6	12	Male	0.8	0.2	302	47	67	6.7	3.5	1.1
7	34	Female	1.2	0.5	180	35	50	6.5	3.1	0.48
8	29	Male	0.8	0.3	200	22	28	6.0	2.9	0.82
9	47	Female	3.4	1.2	210	40	62	7.2	3.5	0.83
10	25	Male	1.0	0.2	120	30	25	5.9	3.0	1.0
11	53	Male	2.1	0.9	300	45	48	6.8	3.2	0.74
12	38	Female	2.5	1.1	500	39	60	6.4	2.5	0.89
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	75	Female	0.8	0.2	188	20	29	4.4	1.8	0.6
2	60	Male	8.9	4.0	950	33	32	6.8	3.1	0.8
3	20	Female	16.7	8.4	200	91	101	6.9	3.5	1.02
4	60	Male	2.0	1.1	664	52	104	6.0	2.1	0.53
5	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
6	32	Male	25.0	13.7	560	41	88	7.9	2.5	2.5
7	28	Female	1.2	0.5	240	22	29	6.9	3.1	1.1
8	72	Male	0.5	0.2	120	15	25	6.8	3.4	0.9
9	38	Female	5.4	1.5	320	45	55	6.4	2.8	0.6
10	50	Male	2.3	1.1	212	30	35	7.1	3.9	1.5
11	55	Female	0.8	0.1	200	25	40	7.3	4.0	0.8
12	33	Male	6.1	2.3	180	40	60	6.7	3.5	1.3
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3